LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur, Pokhara Nepal

A Project Proposal

On

"Simple Banking System"

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Submitted by:

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1.Introduction

A Simple Banking System is a software-based system where users can perform banking activities digitally in an easy and interactive way. The main purpose of this project is to provide users with a simple platform to manage their banking operations while also understanding the concepts of programming, database management, and software development. This system helps to manage user accounts, maintain balance records, process deposits and withdrawals, and store transaction history efficiently.

In the existing traditional method, banking records are often managed manually using registers and paperwork, which can lead to calculation errors, data loss, and difficulties in maintaining records properly. To overcome these problems, this project was selected as it provides a computerized and user-friendly banking environment. The system helps users perform banking tasks smoothly without manual calculation errors and improves the overall banking experience.

This project also helps to reduce manual work and increase accuracy in banking operations. The system can quickly update account balances, record transactions automatically, and provide secure access through login and registration features. Since everything is handled digitally, the chances of mistakes are minimized and the banking process becomes faster, safer, and more reliable.

During the development of this project, different tasks were divided among team members such as documentation, coding, designing flowcharts, testing, and research work. Every member contributed equally to improve the system and make it more effective. Research was also carried out to understand the problems in manual banking systems and to develop suitable solutions using programming concepts.

In today’s digital world, computerized banking systems are more efficient and reliable than manual systems. By developing the Simple Banking System, users can experience a digital banking platform that is fast, accurate, and easy to use. This project also helps beginners improve their programming knowledge, problem-solving skills, and understanding of software development concepts.

2.Aim and Scope

The main aim of the Simple Banking System is to develop a basic and efficient digital banking application using programming concepts. The project helps users understand how real banking systems work while improving logical thinking, problem-solving skills, and software development knowledge. It is also designed to enhance programming skills such as using arrays, functions, loops, conditional statements, file handling, and user input management in the C programming language.

The scope of the Simple Banking System includes basic banking operations such as user registration, login authentication, account creation, balance inquiry, deposit, withdrawal, and transaction history management. The system allows users to securely manage their accounts and perform transactions in a structured and organized manner. It can also be used for educational purposes to help beginners understand the fundamentals of database handling, file management, and system design.

The project can also be expanded in the future by adding features such as:

- Multi-user role system (admin and customer)
- Advanced security features like encryption and OTP verification
- Mini statement and detailed transaction reports
- Loan and interest calculation system
- Graphical user interface (GUI) for better user experience
- Mobile or web-based banking integration
- Budget tracking and financial analytics tools

3.Objective

The importance of project objectives is essential as they define the goals that are achieved during the development of the system. Project objectives are SMART; meaning they are specific, measurable, achievable, relevant, and time-bound. The main focus of the project objectives is the development of a Simple Banking System and its features.

- Development of a simple and efficient banking system application.
- User account management including registration and secure login system.
- Proper handling of banking operations such as deposit and withdrawal.
- Automatic balance calculation and real-time updates of account information.
- Secure transaction processing without errors using programming techniques.
- Use of programming concepts such as loops, arrays, functions, and file handling.
- Maintenance of accurate transaction history for each user.
- Creation of a user-friendly interface for easy navigation and usage.
- Reduction of limitations compared to traditional manual banking systems.
- Improvement of logical thinking and problem-solving skills through project development.
- Successful completion of the project within the given time and available resources.

4.Previous Study

The study of previous banking systems helps developers understand how banking applications were designed, developed, and improved over time. Earlier banking systems were mostly manual or basic computer-based systems where financial transactions such as deposits, withdrawals, and balance checks were recorded using simple methods. These systems focused on core banking operations such as customer account management, transaction processing, and generating basic account statements.

In traditional computer-based banking systems, the software was developed using fundamental programming concepts like loops, arrays, conditional statements, and functions, along with database integration. Most previous systems allowed only basic operations between bank staff and customers, while more advanced versions introduced features such as ATM services, online banking, and automated transaction processing using secure authentication methods.

The study of these systems shows that simplicity, accuracy, and reliability were the key factors that made banking applications effective for users. However, early systems often lacked user-friendly interfaces and strong security mechanisms, which limited their efficiency and user trust.

The previous banking systems also revealed several limitations. Many older systems had poor user interfaces, weak error handling, and lacked features such as transaction history tracking, real-time updates, fund transfer options, and multi-user access control. Some systems also faced issues like data inconsistency and improper validation of user inputs, which affected performance and security.

By studying earlier banking systems, developers can identify strengths and weaknesses and design improved banking applications. The new system can enhance security, improve user experience, provide real-time transaction processing, ensure accurate data management, and include additional features such as mobile banking, notifications, and advanced authentication methods for better reliability and usability.

5.Current Problems and Purposed Solution

Many beginner-level banking systems available today are either too complex for new users or lack proper interaction for customers and trainees learning financial system development. Traditional banking applications often have limited features, weak input validation, unclear transaction displays, and no proper handling of invalid operations. Some existing console-based banking programs may also contain logical errors such as incorrect balance updates, transaction failures, or difficulty restarting or resetting sessions after completion.

In educational environments, students learning banking system development need simple projects that help them understand concepts like arrays, loops, functions, database handling, and conditional statements. However, many available examples are incomplete, difficult to understand, or not user-friendly. This creates difficulty for beginners in learning financial system design and logical problem-solving skills effectively.

To solve the problems found in previous banking systems, a simple and reliable banking management system will be developed using the C programming language. The proposed system will focus on easy transactions, proper input handling, and accurate account processing so that users can manage banking operations without errors or confusion.

The main solutions included in the proposed system are:

- Develop the system with a simple and user-friendly console interface.
- Use a structured account-based system for easy management of banking data.
- Allow users to perform basic operations such as deposit, and balance inquiry.
- Validate user input to prevent invalid transaction entries.
- Prevent operations on non-existent or unauthorized accounts.
- Ensure correct processing of transactions in all cases.
- Detect invalid or duplicate operations automatically.
- Avoid infinite loops and logical programming errors.
- Provide options to continue transactions, restart session, or exit the system.
- Use programming concepts like arrays, loops, functions, and conditional statements properly.

6.Requriment

6.1 Functional Requirements

The Banking Management System is designed to support basic banking operations for customers and administrators. The system should allow users to create bank accounts, deposit money, withdraw funds, and check account balances securely. It must ensure that only valid transactions are processed and prevent operations such as withdrawing more money than the available balance. The system should automatically update account information after every transaction and maintain transaction records accurately. Additionally, the system should display appropriate messages for successful transactions, insufficient balance, invalid account details, and other banking activities.

6.2 Non-Functional Requirements

The system should be designed to be secure, reliable, and user-friendly. It must provide quick responses during banking transactions and ensure accurate handling of customer data. The interface should be simple and easy to understand for both customers and administrators. The system should be maintainable and structured properly so future modifications can be implemented easily. If developed in a programming language like C, Java, or Python, the system should be portable across different operating systems with minimal changes. In addition, the system should efficiently use memory and processing power while maintaining smooth and uninterrupted performance.

7.System Design

In the use case diagram, the customer interacts with the Banking System by performing different banking operations such as creating an account, depositing money, withdrawing money, and checking account balance. Each transaction triggers the system to validate the entered information and update the customer's account details automatically. The system then displays the transaction result, such as successful deposit, withdrawal, balance information, or error messages for invalid operations. This process continues as long as the customer performs banking activities. After completing one transaction, the customer can choose another banking service or exit the system.

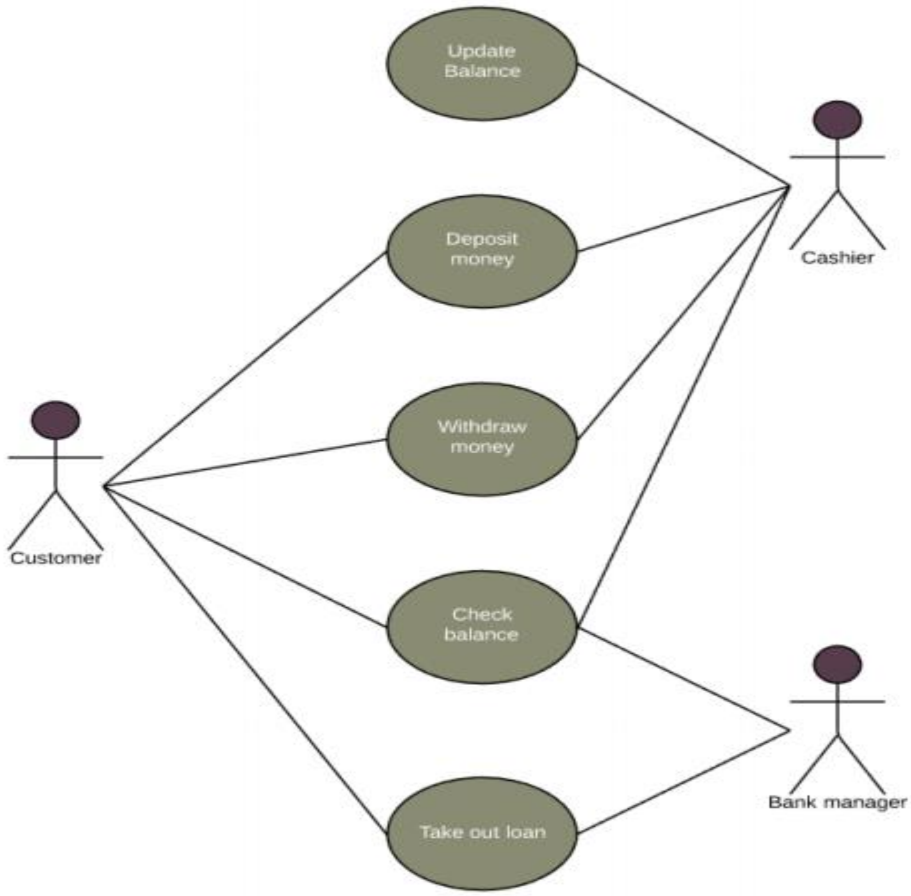

Figure 1: Use Case Diagram

8.Detail Timeline

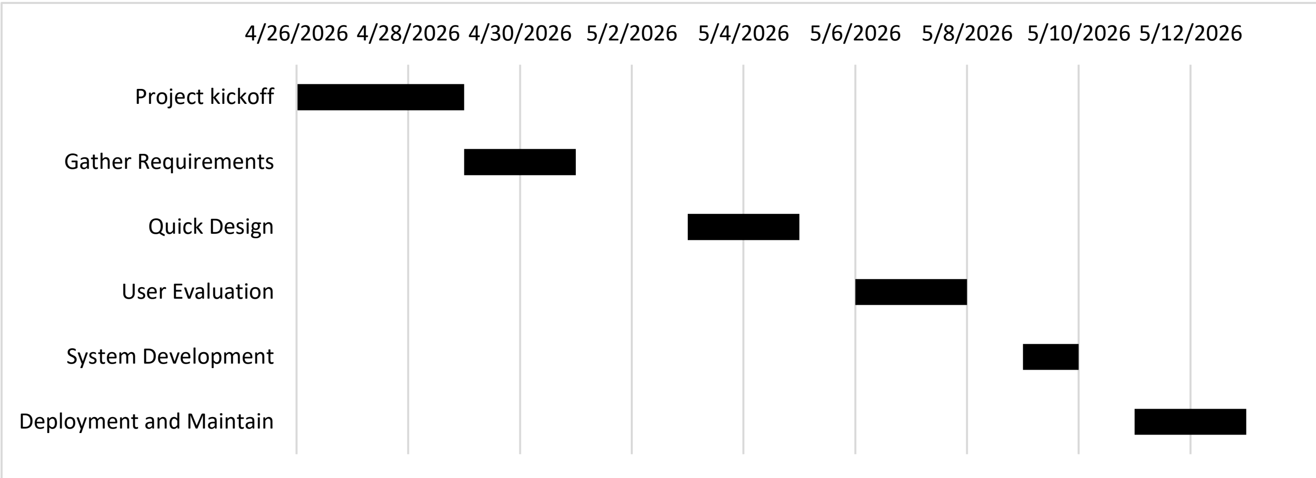

Figure 2 : Gantt Chart

9.Conclusion

In conclusion, the development of the Simple Banking System using the C programming language is a useful and practical project that helps strengthen fundamental programming skills. This project demonstrates the effective use of basic programming concepts such as arrays, loops, conditional statements, functions, file handling, and logical operations, which are essential in structured programming.

The banking system is simple in design but highly effective in improving problem-solving skills and logical thinking. It also provides an understanding of how a real-world application can be designed, implemented, and executed step by step. Through this project, we learn how customer information is managed, how banking transactions are processed, and how operations such as deposits, withdrawals, and balance checking are handled efficiently.

Additionally, this project highlights the importance of planning and maintaining a proper program structure before coding. It also helps in developing debugging and testing skills by identifying and fixing errors during implementation. Although the Simple Banking System is a basic application, it creates a strong foundation for developing more advanced banking and financial management systems in the future.

Overall, this project not only fulfills academic requirements but also builds confidence in programming and enhances our ability to design secure, efficient, and logical solutions using the C programming language.

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